



Leandro Di Bella

PERCEPTION & ROBOTICS ENGINEER | PHYSICAL AI FOR AUTONOMOUS SYSTEMS

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“Deployable autonomous perception with temporal consistency and robustness under distribution shift.”

Summary

PhD candidate looking for a full-time opportunity. Specialized in robotics, physical AI, and computer vision for autonomous systems, with expertise in perception, detection, multi-object tracking, and vision-language models (VLMs). Currently technical lead on drone localization and navigation within the ORIGAMI (EDF) project, and working on multi-agent motion forecasting using flow-based generative AI. Experienced in deploying real-time perception on embedded platforms (Jetson, TensorRT, C++).

Experience

IMEC & VUB (ORIGAMI, EDF project) TECHNICAL LEAD, ROBOTICS & PERCEPTION RESEARCHER	Jan. 2026 – Present Brussels, Belgium	Technical lead for drone autonomy: architected and integrated the full drone platform (hardware, sensing, and autonomy software). Sensor fusion and localization in 3D environments; developing world-model-based methods for autonomous navigation.
MACQ Mobility (research project) COMPUTER VISION RESEARCHER	Jan. 2023 – Jan. 2026 Brussels, Belgium	Detection, 3D multi-object tracking, scene understanding, and motion forecasting for autonomous vehicles using generative AI. Achieved #1 on the KITTI MOT and Waymo motion forecasting benchmarks.
Mappx FOUNDER	Aug. 2025 – Present Brussels, Belgium	Built a map-centric social travel app (Flutter) with FastAPI backend and PostgreSQL.
MACQ Mobility COMPUTER VISION INTERN	Aug. 2022 – Oct. 2022 Brussels, Belgium	Developed and integrated instance segmentation on NVIDIA Jetson TX2 (Python/C++).
Vrije Universiteit Brussel TEACHING ASSISTANT	2022 – 2025 Brussels, Belgium	Teaching assistant for Machine Learning and Big Data Processing (2.5 years).

Education

ETRO, Vrije Universiteit Brussel (VUB) PhD in Engineering Sciences (AI, Computer Vision & Robotics)	Brussels, Belgium 2023 – Present
ICVSS 2026 (International Computer Vision Summer School) Summer school: Computer Vision for Spatial and Physical Intelligence	Sicily, Italy 2026
Solvay Brussels School Advanced Master in Industrial & Technological Management	Brussels, Belgium 2025 – Present
Bruface (ULB/VUB) M.Sc. Electrical Engineering (Information Technology Systems)	Brussels, Belgium 2020 – 2023

Selected publications

HybridTrack: A Hybrid Approach for Robust Multi-Object Tracking L. Di Bella, Y. Lyu, B. Cornelis, A. Munteanu IEEE Robotics and Automation Letters (RA-L) / ICRA (2025). Ranked #1 on the KITTI MOT benchmark.	ReferGPT: Towards Zero-Shot Referring Multi-Object Tracking T. Chamiti, L. Di Bella, A. Munteanu, N. Deligiannis CVPR Workshops (2025)
FlowS: One-Step Motion Prediction via Local Transport Conditioning L. Di Bella, A. Munteanu, B. Cornelis IEEE Robotics and Automation Letters (RA-L), accepted (2026). Ranked #1 on the Waymo motion forecasting benchmark.	Spectral-Aware Multi-Object Tracking in Harsh Aerial Perception Domains L. Di Bella, J. Mimassi, L. Denis, A. Munteanu IEEE ICRA 2026 Workshop S2S

Research interests

Perception for autonomous systems, world models for perception and navigation, localization in 3D environments, vision-language-action (VLA) models for robotics.

Skills

Programming	Python, C++, CUDA, PyTorch, TensorRT, ONNX, FastAPI, Flutter/Dart
ML / Perception	Detection, instance segmentation, monocular 3D, 2D/3D tracking (MOT), SLAM, Kalman filtering, temporal consistency
Vision-language & GenAI	Grounding / referring tracking, multimodal reasoning, generative motion forecasting
Robotics	ROS 2, SLAM, localization, motion planning, navigation, sensor fusion
Systems / Deployment	Jetson, Docker, GitHub CI/CD, Azure cloud services, Firebase, PostgreSQL
Academic service	Reviewer: TIP, RA-L, ICRA, ICIP, IET